

Supplementary Information for “Resistance-Aware Computational Prioritization of Antimalarial Leads from African Natural Products: Docking, Chemical-Space Descriptors, and Exploratory Molecular Dynamics”

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Table S1: Docking protocol validation: enrichment metrics and re-docking accuracy. Values reproduce the corrected benchmarks from the parent study (V2-corrected grids; AutoDock Vina + DiffDock consensus). MMV Malaria Box values are consensus-score ROC-AUCs for the three targets with MMV docking data (PfDHFR, PfCRT, PfATP4); DEKOIS 2.0 validation used AutoDock Vina only for PfDHFR. BEDROC values use $\alpha = 20$ (MMV). Re-docking RMSD is the heavy-atom RMSD between the top-scoring docked pose and the co-crystallised ligand; MTX redocking failed (RMSD ≈ 30 Å).

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC	RMSD (Å)
PfDHFR (7F3Y)	Vina redock	–	–	–	–	N/A
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.309	–
	DEKOIS (Vina only)	0.450	–	0.00	–	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.434	–
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810	–
PfClpR (4GM2; not PfClpP)	MMV enrichment	not evaluated	–	–	–	–

Identity note. PDB 4GM2 is PfClpR rather than PfClpP; it is not used as PfClpP validation.

Table S2: Endpoint-analysis interpretation for the four parent-study consensus complexes used for production MD. These systems are distinct from the 17 set-C polypharmacology candidates. Mean $\pm$ standard deviation is shown only for the single interpretable MM-GBSA result from 101 snapshots; dissociated or conversion-corrupted systems are reported as N/A. Mutant RRS is a docking-based analysis of set C and is not assigned to these MD systems.

Ligand	Target	ΔG_{bind}	Std. dev.	Interpretation
164	PfClpR-labelled cohort*-WT	–	–	N/A; ligand dissociated [‡]
201	PfDHFR-WT	–	–	N/A; ligand dissociated [‡]
214	PfCRT-WT	–18.25	0.40	Interpretable (gmx_MMPBSA)
438	PfATP4-WT	–	–	N/A; conversion corrupted [§]

*PDB 4GM2 is PfClpR rather than PfClpP; the 164-labelled system is therefore not structural validation of PfClpP.

[‡]Ligand dissociated during production; no binding free energy is inferred.

[§]Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion.

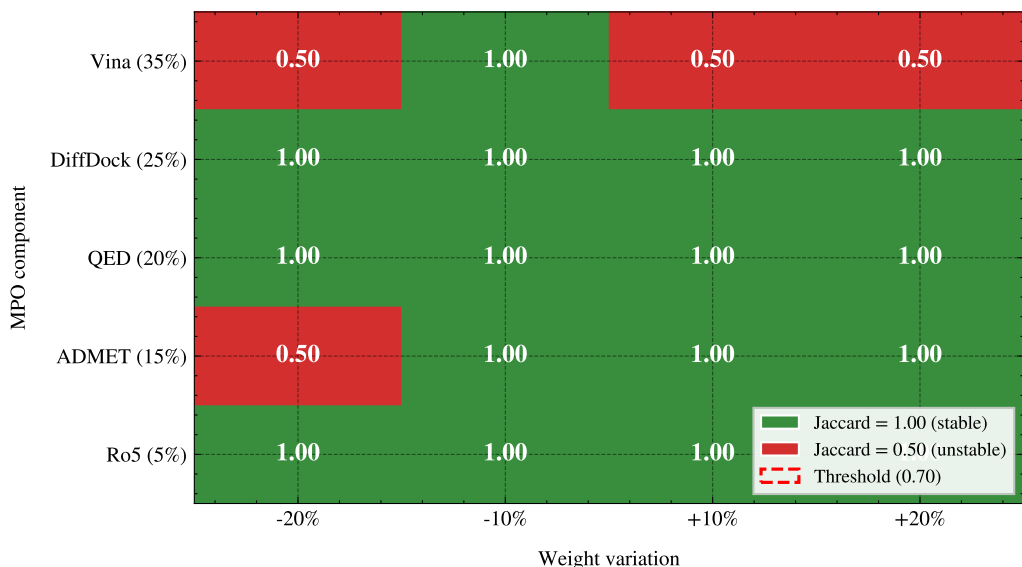


Figure S1: MPO sensitivity analysis heatmap. Each cell shows the Jaccard similarity of the top-20 hit list when the corresponding MPO weight is varied by $\pm 20\%$. Green (1.00) indicates the hit list is unchanged; red (0.50) indicates substantial reordering.

Table S3: Predicted ADMET profile of the 17 set-C polypharmacology candidates (ADMET-AI, parent-library screening). All values are machine-learning predictions; experimental ADMET profiling and three-tool concordance were outside the scope of this study. Range across candidates: LogS [0.34, 0.73], CYP3A4 inhibition ≥ 0.5 for 7/17 candidates, Caco-2 permeability ≥ 0.5 for 15/17, Clint_h [0.48, 0.63].

Compound	ADMET-AI LogS	CYP3A4 prob.	Caco-2 prob.	Clint _h
PP-01	0.34	0.39	0.63	0.60
PP-02	0.46	0.97	0.86	0.55
PP-03	0.43	0.62	0.95	0.55
PP-04	0.40	0.58	0.92	0.63
PP-05	0.61	1.00	0.35	0.51
PP-06	0.54	0.90	0.97	0.60
PP-07	0.36	0.02	0.75	0.62
PP-08	0.59	1.00	0.41	0.56
PP-09	0.65	0.13	0.81	0.57
PP-10	0.49	0.13	0.93	0.62
PP-11	0.39	0.06	0.96	0.58
PP-12	0.49	0.00	0.98	0.48
PP-13	0.57	0.97	0.98	0.53
PP-14	0.66	0.05	0.68	0.49
PP-15	0.66	0.49	0.95	0.51
PP-16	0.54	0.25	0.91	0.50
PP-17	0.73	0.01	0.82	0.50

Table S4: Candidate-cohort ACSI weight sensitivity. One ACSI component weight was varied by $\pm 20\%$ and the remaining weights were rescaled proportionally to preserve a unit sum. Mean and standard deviation refer to the resulting ACSI values across the 17 candidates. Rank stability is reported as Spearman correlation over all candidates and Jaccard overlap of the top five candidates with the baseline ranking. Because the full-library component matrix was not part of the committed source-data record, this table is a local sensitivity analysis of the canonical Set-C cohort and does not establish robustness of a full-library top-20 ranking.

Weight varied	Variation (%)	New weight	Mean ACSI	SD	Spearman ρ	Top-5 Jaccard
D_{DrugBank}	-20	0.320	0.512	0.149	0.9779	0.6667
D_{DrugBank}	+20	0.480	0.574	0.159	0.9338	0.6667
D_{ANPDB}	-20	0.200	0.568	0.149	0.9167	0.6667
D_{ANPDB}	+20	0.300	0.518	0.158	0.9804	0.6667
f_{sp^3}	-20	0.160	0.552	0.152	0.9510	0.6667
f_{sp^3}	+20	0.240	0.534	0.153	0.9583	0.6667
NPL	-20	0.120	0.535	0.160	0.9632	0.6667
NPL	+20	0.180	0.551	0.145	0.9804	1.0000

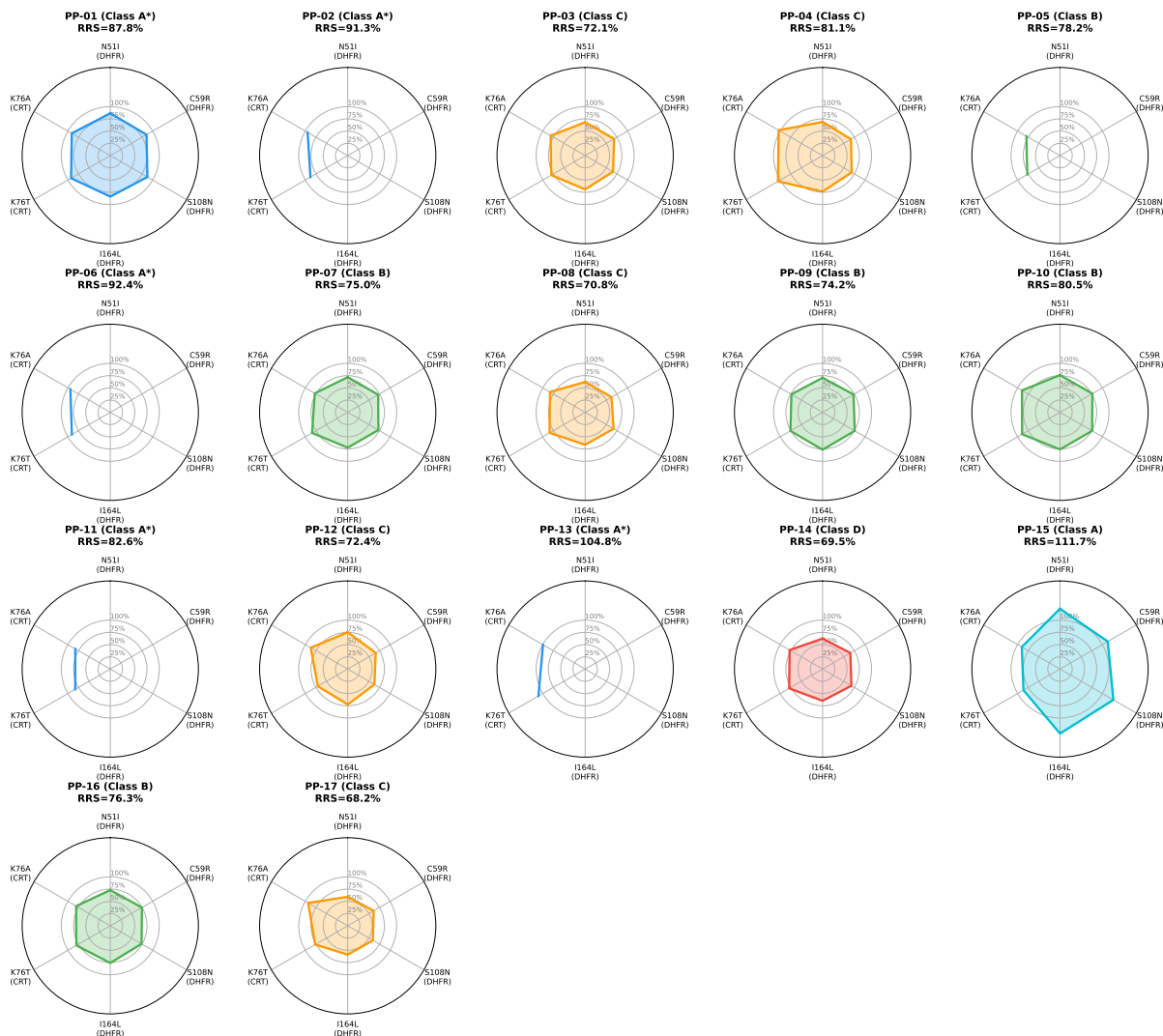


Figure S2: Per-compound, docking-derived RRS profiles across six resistance mutations for the 17 Set-C candidates. Missing axes denote an ineligible target under the 5.0 kcal mol⁻¹ wild-type docking-score threshold and are not treated as zero. Concentric rings indicate 50 %, 100 %, and 150 % RRS; the plot does not encode absolute binding potency.

Scope of the MD evidence. The parent-study trajectories provide a targeted, non-membrane explicit-solvent check on four wild-type complexes; they are distinct from the separate Set-C production pilot and do not constitute a physiologically complete model for the membrane targets. The authoritative post-production chain is the completed QC output recorded in the pilot manifest; all 16 pilot systems passed the declared gate. A 16-system Set-C MD pilot (PP-01 and PP-02 against the six-mutant panel, 10 ns each) passed the declared trajectory-QC gate and is reported as a secondary analysis; its trajectory-derived MD-RRS and mutant-state MM-GBSA estimates are summarized in Table S8 and compared with docking RRS in Figure S3.

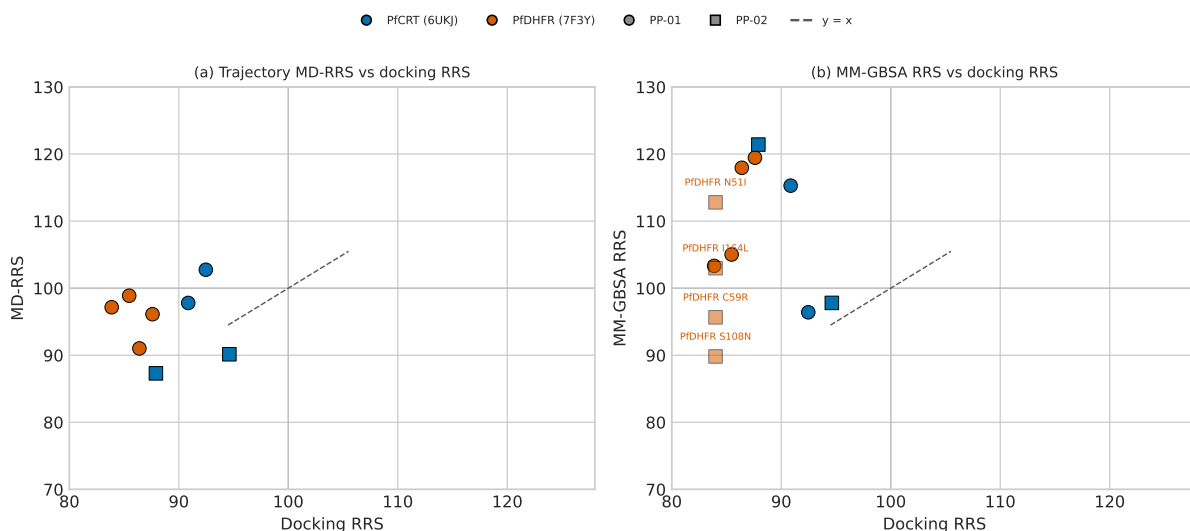


Figure S3: Set-C MD pilot trajectory-derived RRS estimates versus docking RRS (WT-normalized, WT = 100). Panel (a): MD-RRS (mean-minimum-distance geometry) versus docking RRS for the eight mutant systems with both estimates; panel (b): MM-GBSA RRS versus docking RRS, with the four PP-02 PfDHFR systems (no docking WT reference in the pilot) shown on the MM-GBSA axis only. Points are coloured by target (PfCRT, blue; PfDHFR, orange) and shaped by candidate (PP-01, circle; PP-02, square). For the eight mutant systems with a docking WT reference, docking predicted looser mutant binding (RRS < 100); the trajectory-derived estimates do not reproduce this pattern among the comparable systems. Secondary, exploratory analysis; no primary claim is based on it.

The stable identifiers were retrieved from PlasmoDB/VEuPathDB annotations on 12 August 2026 (PlasmoDB Consortium, 2026); record-level permalinks and the retrieval date are preserved in the machine-readable CSV. The exact database release/build was not captured in the source snapshot and is therefore not treated as an inferred annotation version. The existing PfCRT pH 5.2 redocking sensitivity (100/100 library ligands; $\rho = 0.270$, $p = 0.006551$) is documented separately in the project audit and is not merged into the Set-C docking-RRS cohort. Because the present study does not provide an independent multi-gene targetome or a prespecified parasite-proteome background, no GO or pathway-enrichment test was performed. This table therefore supplies biological identity and mutation provenance without converting docking scores into pathway-level evidence (Trapotsi et al., 2022).

Table S5: Stable *Plasmodium falciparum* target identifiers and mutation context used in this study. Identifiers are reported for annotation and provenance only; they do not constitute pathway-level target engagement, causal network evidence, or experimental validation. The accompanying source-data record preserves the target annotations and retrieval provenance.

Target	Stable gene ID	Structural context	Mutation or evidence boundary
PfDHFR	PF3D7_0417200	7F3Y WT template	N51I, C59R, S108N, I164L; antifolate-resistance states
PfCRT	PF3D7_0709000	6UKJ WT template	K76T clinical resistance marker; K76A comparator state
PfATP4	PF3D7_1211900	9N10 parent-study structure	Target annotation only; not part of the Set-C mutation panel
PfClpP	PF3D7_0307400	Proteolytic Clp subunit	Not assigned to PDB 4GM2; no PfClpP claim is inferred from the 4GM2 record
PfClpR	PF3D7_1436800	4GM2, PfClpR(P61–E244)	4GM2 is PfClpR, not PfClpP; parent-study identity retained explicitly

Table S6: Cohort and estimand separation. The Set-C docking cohort contains 17 candidates and 136 WT/mutant systems; 12 candidates have complete eligible PfDHFR/PfCRT panels and define the target-balanced primary RRS analysis, whereas 5 are PfCRT-only sensitivity records. The parent-study MD cohort is non-overlapping. The Set-C MD pilot passed the declared trajectory-QC gate (16/16) and is reported as a secondary, exploratory analysis whose MD-RRS and MM-GBSA estimates are not merged into the primary docking-RRS classification.

Evidence layer	Cohort	Systems	Primary analyses	Interpretation boundary
Set-C docking	PP-01–PP-17	136 (12 complete; 5 PfCRT-only)	WT/mutant docking; target-specific RRS, ACSI, PNS	Docking-derived prioritisation; not measured binding or MD-RRS
Parent-study MD	Ligands 164, 201, 214, 438	4 WT complexes $\times$ 10 ns	RMSD, contacts, bound-state assessment, MM-GBSA where valid	Separate structural check; one interpretable MM-GBSA estimate (PfCRT–214)
Set-C MD pilot	PP-01/PP-02	16 systems (16/16 QC PASS)	Trajectory QC, discriminative MD-RRS, MM-GBSA	Reported as secondary; not merged into primary RRS classification
Full Set-C MD-RRS	PP-01–PP-17	136 planned systems	Full trajectory QC and MD-RRS	Not computed; requires complete PASS-QC cohort

Table S7: Sensitivity of the PNS ranking to the imputed PfCRT centrality. The canonical value is the observed network-mean imputation ($C_{\text{PfCRT}} = 0.151$). Alternative values are zero, one-half, one-and-a-half, and twice the canonical value. Spearman ρ is calculated against the canonical 17-candidate ranking; all candidates have two target entries in the PNS record. This analysis assesses ranking sensitivity to an unresolved network-data choice and does not establish target essentiality or biological polypharmacology.

Imputation	C_{PfCRT}	Spearman ρ	Minimum PNS	Maximum PNS	n
Zero	0.0000	0.9716	0.300	5.147	17
One-half canonical	0.0755	0.9975	0.674	5.573	17
Canonical network mean	0.1510	1.0000	1.047	6.000	17
One-and-a-half canonical	0.2265	0.9755	1.421	6.426	17
Twice canonical	0.3020	0.9632	1.795	6.853	17

Secondary calibration and ranking analyses

Tartarus docking calibration

The Tartarus analysis used QuickVina on 19 913 primary leads against 1SYH, 6Y2F, and 4LDE. The composite MPO comparison used up to 17 211 complete observations. The composite association was $\rho = 0.0129$, $p = 0.0913$; target-specific associations were small and are interpreted as exploratory rather than as evidence of independence or score equivalence.

Table S8: Tartarus docking calibration: Spearman correlations between QuickVina scores and weighted MPO scores.

Score	Spearman ρ	p -value	n	Interpretation
Composite (mean of 3 targets)	0.0129	9.13×10^{-2}	17 211	No meaningful association
1SYH (PfDHFR)	0.0655	8.18×10^{-18}	17 205	Small effect
6Y2F (PfCRT)	-0.0040	6.04×10^{-1}	17 211	Null
4LDE (A2A receptor)	-0.1415	1.17×10^{-77}	17 211	Small effect

Tartarus cross-docking correlation

Only 24 molecules overlapped between the Tartarus and project docking records. The structure-mismatched comparison is exploratory and cannot establish experimental calibration.

Table S9: Tartarus independent-tool comparison for the 24-molecule overlap.

Tartarus	Project metric	n	ρ	p	Interpretation
Composite	4GM2/PfClpR	24	0.246	2.47×10^{-1}	No
Composite	6UKJ/PfCRT	24	0.075	7.28×10^{-1}	No
Composite	7F3Y/PfDHFR	24	-0.110	6.10×10^{-1}	No
Composite	9N10/PfATP4	24	0.118	5.83×10^{-1}	No
1SYH/DHFR	7F3Y/PfDHFR	24	-0.442	3.04×10^{-2}	Exploratory

Complete ACSI components

ACSI was calculated as

$$\text{ACSI} = 0.40D_{\text{DrugBank}} + 0.25D_{\text{ANPDB}} + 0.20f_{sp^3} + 0.15\text{NPL}.$$

Components were min-max normalized over the selected 17-candidate cohort. Two candidates exceeded 0.70, the cohort mean was 0.543, and the complete component values are retained in the machine-readable source table. The eight one-at-a-time $\pm 20\%$ perturbations gave Spearman rank correlations of 0.9167–0.9804 and top-five Jaccard overlaps of 0.6667–1.0000.

PNS ranking and imputation

The STRING network used confidence threshold 700. PfCRT centrality was imputed with the network mean, 0.151. The complete 17-candidate ranking is retained in the source table. Imputation sensitivity over zero, half, canonical, one-and-a-half, and double canonical centrality yielded rank correlations of 0.9632–1.0000. This is sensitivity to a network-data choice, not biological validation.

Table S10: PNS ranking and RRS class. PfCRT-only denotes candidates without an eligible PfDHFR WT denominator for primary RRS.

Candidate	PNS	ACSI	RRS class
Rank 1	6.00	0.599	C
Rank 2	4.71	0.308	C
Rank 3	4.53	0.567	C
Rank 4	4.48	0.603	B
Rank 5	4.48	0.551	B
Rank 6	4.45	0.173	B
Rank 7	4.45	0.459	A*
Rank 8	4.36	0.743	D
Rank 9	4.36	0.605	C
Rank 10	4.34	0.594	C
Rank 11	4.20	0.601	B
Rank 12	3.50	0.576	A
Rank 13	2.32	0.597	PfCRT-only
Rank 14	1.23	0.823	PfCRT-only
Rank 15	1.14	0.387	PfCRT-only
Rank 16	1.10	0.458	PfCRT-only
Rank 17	1.04	0.589	PfCRT-only

Predicted ADMET

The complete ADMET-AI endpoint table for all 17 candidates is provided in table S3. These are single-source machine-learning predictions without experimental validation or inferred inter-tool concordance.

References

- PlasmoDB Consortium. PlasmoDB: The *Plasmodium* genomic resource. <https://plasmodb.org/>, 2026.
- Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (moa): data, methods and integration. *RSC Chemical Biology*, 3(2): 170–200, 2022. doi: 10.1039/D1CB00069A.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.

Table S11: Set-C MD pilot endpoint estimates (secondary, exploratory analysis). All 16 systems passed the declared trajectory-QC gate; ΔG_{bind} is an MM-GBSA endpoint estimate averaged over 100 snapshots with within-trajectory standard deviation, not a replicate uncertainty or a thermodynamic confidence interval. MMG-RRS is the ratio of absolute endpoint magnitudes relative to the corresponding WT system (WT = 100); it is not a free-energy resilience measurement. n/a = no WT docking reference for PP-02 PfDHFR. The two systems flagged with * failed the first MM-GBSA pass (BOND/UB overflow from a protein split across the periodic boundary) and were re-run on PBC-whole trajectories. MMG-RRS values above 100 % carry the same interpretation as their docking counterparts: within endpoint noise they indicate retention of wild-type binding, not a gain; eight of the twelve mutant systems lie above this threshold.

Candidate	Target	State	ΔG_{bind} (kcal mol ⁻¹)	SD	MMG-RRS	MD-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	97.8	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	102.8	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	100.0	-
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	98.9	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	97.2	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	91.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	96.1	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	100.0	-
PP-02	PfCRT	K76A*	-24.53	1.27	97.8	90.2	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.3	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	100.0	-
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	102.0	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	97.4	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	72.2	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	97.5	n/a
PP-02	PfDHFR	WT*	-30.16	2.06	100.0	100.0	-